

Option Pricing: PDE vs Monte Carlo

Folder: Option_pricing_MC_BS | Black-Scholes, American put, local volatility, Dupire and Heston

1. Objective

Price options two independent ways - by solving the pricing PDE with a high-order spectral solver, and by Monte Carlo simulation (via Feynman-Kac) - and use each to cross-check the other. The folder progresses from vanilla Black-Scholes to American exercise, local volatility, Dupire's forward equation and the two-dimensional Heston model.

Shared numerical engine. Every PDE notebook uses the same spectral method-of-lines machinery: the solution is expanded in Chebyshev polynomials ($n = 50$ modes; the differentiation matrix is loaded from `diffmatrix100.txt`), the PDE is enforced at Gauss-Lobatto collocation nodes, and boundary conditions replace rows of the collocation system. Time is then advanced with an 8-stage singly diagonally implicit Runge-Kutta (SDIRK/DIRK) scheme at a step of 10^{-3} .

Domains are split into two Chebyshev cells joined by continuity of value and slope; the log-moneyness coordinate is $x = \ln(S/K)$ on $[-w, w]$, time is $\tau = T - t$. A shared `diffmatrix100.txt` sits at the folder root.

2. european_call/

- **european_call_option.ipynb** - Black-Scholes PDE $d_{\tau} f = \frac{1}{2} \sigma^2 f_{xx} + (r - \sigma^2/2) f_x - r f$ with $r = 5\%$, $\sigma = 20\%$, $K = 1$, $T = 1$, $w = 1.2$; payoff $\max(K e^x - K, 0)$, $f = 0$ at the left edge and $K(e^w - e^{-r\tau})$ at the right. Compared with the analytical formula.
- **european_call_option.cpp / .exe** - the same finite-difference scheme re-implemented in C++, writing the solution grid to `european_call_option_data.txt`.
- **european_call_option_visualize.ipynb** - loads the C++ output, plots the price surface and the log-error heatmap against the analytical call.
- **monte_carlo_european_call_option.ipynb** - GBM path simulation ($S_0 = 1$, $r = 3\%$, $q = 1\%$, $\sigma = 20\%$); sample sizes from 10^2 to 10^6 walkers compared with Black-Scholes, with the observed error against the estimated standard error.

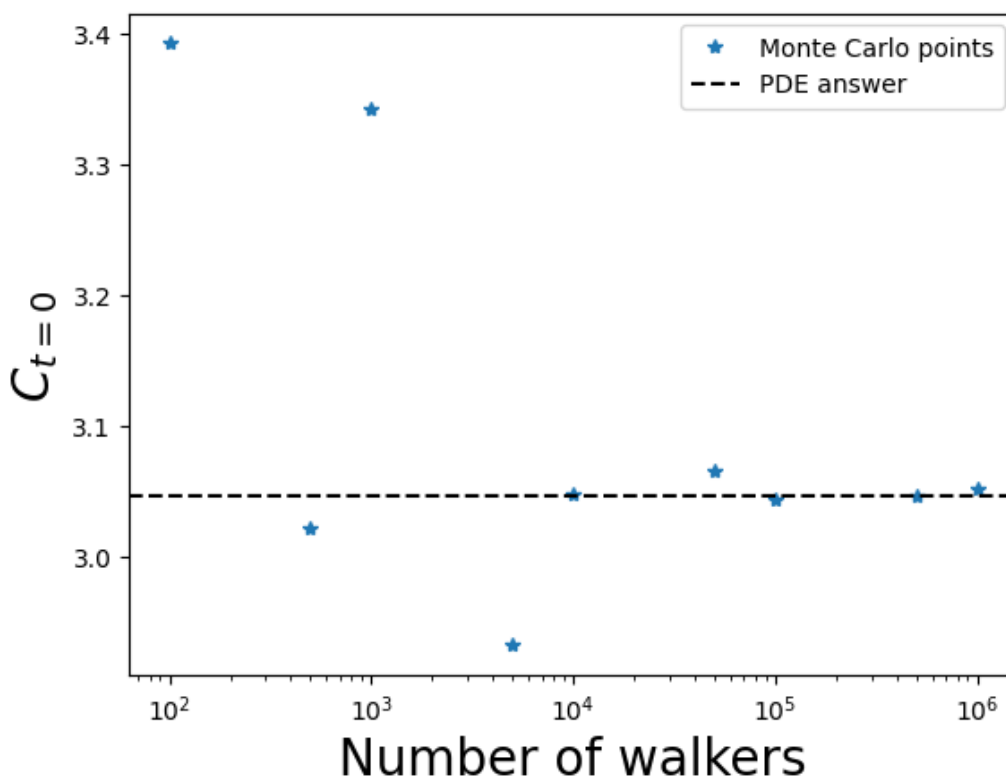
3. american_put/

- **american_put_option.ipynb** - handles early exercise with a penalty method: the PDE gets an added term $\lambda * \max(0, K - K e^x - f)$ with $\lambda = 10^4 r$, enforcing $f \geq$ intrinsic value. Same spectral solver, $r = 5\%$, $\sigma = 20\%$.
- **monte_carlo_american_put_option.ipynb** - Longstaff-Schwartz least-squares Monte Carlo (polynomial degree 3). It first validates the path generator on a European put, then tests convergence against the PDE price (0.06086) as the number of paths N and exercise dates M vary. The notebook concludes that the number of time steps, rather than walkers, is the limiting factor.
- **american_put_montecarlo.gif** - animation of the Monte Carlo exercise process.

4. forward_backward_calibrated_vol/

Treats a European call ($S_0 = 100$, $K = 120$, $r = 3\%$, $q = 1\%$, $T = 1$) under a state- and time-dependent **local volatility** surface $\sigma_{LV}(S, t)$, parametrised by $(\sigma_0, \beta, \kappa, \eta, \lambda) = (0.20, 0.45, 1.5, 0.35, 2.0)$. Setting $\beta = \eta = 0$ reduces it to constant volatility.

- **black_scholes_backward.ipynb** - backward Black-Scholes PDE with $\sigma_{LV}(S, t)$; gives one price for one strike.
- **dupire_forward.ipynb** - Dupire forward PDE in (K, T) : $d_T C = 1/2 \sigma^2 K^2 C_{KK} - (r - q) K C_K - q C$, transformed to $x = \ln(K/S_0)$. One forward solve returns prices for all strikes and maturities at once.
- **feynman_kac_monte_carlo.ipynb** - simulates the same local-vol SDE with up to 10^6 walkers and compares against the PDE answer (about 3.0475) on log axes.
- **blackscholes_pde.gif**, **dupire_pde.gif**, **monte_carlo_plot.png** - visual outputs.



Monte Carlo convergence to the PDE price (monte_carlo_plot.png).

5. heston_model/

Stochastic volatility, so the PDE becomes two-dimensional in (x, v) :

$$dS = (r - q) S dt + \sqrt{v} S dW_1 \quad dv = \kappa (\theta - v) dt + \xi \sqrt{v} dW_2, \\ \text{corr}(dW_1, dW_2) = \rho$$

- Parameters: $\kappa = 2$, $\theta = 0.04$, $\xi = 0.3$, $\rho = -0.7$, $r = 3\%$, $q = 1\%$, $K = 100$, $T = 1$, $w = 2$, $v_{\max} = 1$.
- Tensor-product Chebyshev grid of 50×25 modes per cell (built with a Kronecker product), two cells in x , with a zero-flux condition $dU/dv = 0$ at $v_{\max}$, solved with the same SDIRK integrator. The PDE includes the mixed-derivative term $\rho \xi v U_{xv}$.
- The notebook notes the result was cross-checked with a Fourier-integral reference calculation, and builds a heatmap movie (`heston_pde.gif`) of log call price over (x, v) as time evolves.

6. Takeaways

- Spectral PDE solvers reach very small errors on smooth problems, as seen from the log-error heatmaps and the fast coefficient decay.
- Monte Carlo converges at the expected $N^{-1/2}$ rate and serves as an independent check, particularly in settings (local vol, American exercise) where no closed form exists.
- The Dupire forward equation prices a whole strike/maturity grid in one solve, which is the practical route for calibrating volatility surfaces.

7. File map

Subfolder	Contents
european_call	PDE notebook, C++ solver (.cpp/.exe) and its data file, visualisation, Monte Carlo
american_put	Penalty-method PDE, Longstaff-Schwartz Monte Carlo, GIF
forward_backward_calibrated_vol	Backward BS and Dupire forward with local vol, Feynman-Kac Monte Carlo, GIFs, plot
heston_model	2-D Heston PDE notebook and heatmap GIF